

Circuit name	-	Reference.Pin	-	Connected Reference.Pin	-	Signal Name
IR Signal	-	IRS.1	-	GND	-	
IR Signal	-	IRS.2	-	6volts	-	
IR Signal	-	IRS.3	-	A1.2	-	(IRS Signal
Lights Out	-	A1.3	-	R3.2	-	T2 control signal
Lights Out	-	R3.1	-	T2.2	-	T2 control signal
Lights Out	-	GND	-	T2.1	-	
Lights Out	-	T2.3	-	R2.2	-	+Vtest On/Off
	-		-		-	
12 to 6V Drop Down	-	12V	-	R5.2	-	
12 to 6V Drop Down	-	R5.1	-	R4.2	-	6volts
12 to 6V Drop Down	-	R5.1	-	R1.1	-	6volts
12 to 6V Drop Down	-	R5.1	-	R6.1	-	6volts
12 to 6V Drop Down	-	R4.1	-	GND	-	
+Vtest	-	R1.1	-	R5.1	-	6volts
+Vtest	-	R1.2	-	R2.1	-	+Vtest
+Vtest	-	R1.2	-	EMS1.87a	-	+Vtest
+Vtest	-	R1.2	-	U1.3	-	+Vtest
+Vtest	-	R2.2	-	T2.3	-	+Vtest On/Off
-Vref	-	R8.2	-	GND	-	
-Vref	-	R8.1	-	R7.2	-	
-Vref	-	R7.1	-	U1.2	-	-Vref
-Vref	-	R7.1	-	R6.2	-	-Vref
-Vref	-	R7.1	-	T3.3	-	-Vref
-Vref	-	R6.1	-	R5.1	-	6volts

Circuit name	-	Reference.Pin	-	Connected Reference.Pin	-	Signal Name
Lights On	-	T3.3	-	R7.1	-	-Vref
Lights On	-	T3.2	-	R11.2	-	T3 control signal
Lights On	-	R11.1	-	A1.4	-	T3 control signal
Lights On	-	T3.1	-	GND	-	
+Vtest Voltage Monitor	-	U1.8	-	12volts	-	
+Vtest Voltage Monitor	-	U1.4	-	GND	-	
+Vtest Voltage Monitor	-	U1.3	-	R2.1	-	+Vtest
+Vtest Voltage Monitor	-	U1.2	-	R7.1	-	-Vref
+Vtest Voltage Monitor	-	U1.1	-	L1.1	-	Output 1
+Vtest Voltage Monitor	-	U1.1	-	R10.2	-	Output 1
Output Monitor	-	L1.1	-	R10.2	-	Output 1
Output Monitor	-	L1.2	-	R9.2	-	Output 1
Output Monitor	-	L1.2	-	A1.5	-	Output 1
Output Monitor	-	R9.1	-	GND	-	
EMS Control	-	U1.1	-	R10.2	-	Output 1
EMS Control	-	R10.1	-	T1.2	-	T1 control signal
EMS Control	-	T1.1	-	GND	-	
EMS Control	-	T1.3	-	D1.1	-	
EMS Control	-	T1.3	-	EMS.85	-	Drain
EMS Control	-	D1.2	-	EMS.86	-	EMS On/Off
EMS Control	-	D1.2	-	12volts	-	
EMS wiring	-	12 volts	-	EMS1.87	-	12 volts
EMS wiring	-	EMS1.30	-	Wall Switch	-	

Circuit name	-	Reference.Pin	-	Connected Reference.Pin	-	Signal Name
Common Ground	-	A1.GND	-	Circuit Board GND	-	common ground

Note: If you do not connect the grounds between the Arduino and the Circuit Board, you will have a floating ground and they will not be able to interact with one another